

Lab 5 - Malware Triage

Microsoft Defender Detection and Wazuh SIEM Investigation

Lab Type	SOC / Malware Triage
Test Sample	EICAR antivirus test file
Endpoint	Windows 11 Pro
EDR/AV	Microsoft Defender Antivirus
SIEM	Wazuh 4.14.7
Wazuh Agent	ID 002 / DESKTOP-6CJNNAI

1. Objective

The objective of this lab was to perform a safe malware-triage workflow using the EICAR antivirus test file. The lab demonstrated file identification, SHA-256 hashing, Microsoft Defender detection and quarantine, Windows Defender event analysis, Wazuh SIEM ingestion, and analyst interpretation.

Safety note: EICAR is an industry-standard antivirus test string and is not real malware. It is designed to trigger antivirus products safely for testing purposes.

2. Lab Environment

Component	Configuration
Virtualization	VMware Workstation
Windows endpoint	Windows 11 Pro
Endpoint IP observed in Wazuh	192.168.180.130
Wazuh agent	ID 002 - DESKTOP-6CJNNAI
Wazuh manager/dashboard	192.168.180.129
Defender event channel	Microsoft-Windows-Windows Defender/Operational

3. Test Sample Creation

A safe EICAR test file named **eicar.com.txt** was created in **C:\MalwareLab**. Microsoft Defender immediately detected the file and removed it from the folder.

4. Microsoft Defender Detection

Field	Observed value
Threat name	Virus:DOS/EICAR_Test_File
Severity	Severe
Status	Quarantined
Affected item	C:\MalwareLab\ecar.com.txt
Detection source	Real-Time Protection

Field	Observed value
Origin	Local machine

5. Hash Analysis

The SHA-256 hash of the exact EICAR test content was calculated in PowerShell. Hashing provides a consistent file fingerprint that can be used to identify artifacts across endpoints and security tools.

SHA-256:

275A021BBFB6489E54D471899F7DB9D1663FC695EC2FE2A2C4538AABF651FD0F

6. Windows Defender Event Logging

PowerShell confirmed that Microsoft Defender recorded the detection as Windows Defender Operational Event ID **1116**. The Wazuh agent was then configured to collect the Microsoft-Windows-Windows Defender/Operational event channel.

7. Wazuh SIEM Detection

Field	Observed value	Analyst meaning
Event ID	1116	Microsoft Defender malware detection event
Threat name	Virus:DOS/EICAR_Test_File	Known safe antivirus test signature
Category	Virus	Defender classification
Severity	Severe	High endpoint severity
Path	C:\Malware\aleicar.com.txt	Detected file location
Process	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	PowerShell created the test file
Detection source	Real-Time Protection	Detected during real-time monitoring
Origin	Local machine	Detection occurred on endpoint
Wazuh rule ID	62123	Matched Windows Defender rule
Wazuh rule level	12	High-severity Wazuh alert

8. SOC Analyst Interpretation

Microsoft Defender detected the EICAR test sample through real-time protection and quarantined the file. The corresponding Defender Operational Event ID 1116 was ingested into Wazuh and generated a rule level 12 alert.

Because EICAR is a non-malicious test signature, the alert was expected and no real malware infection occurred. However, the workflow accurately demonstrated how a SOC analyst can confirm endpoint detection, identify the affected file, review the creating process, validate severity, and correlate endpoint telemetry in a SIEM.

9. Investigation Workflow

Step	Action
1	Create safe EICAR test sample
2	Observe Microsoft Defender detection
3	Confirm quarantine in Protection History
4	Calculate SHA-256 hash

Step	Action
5	Verify Defender Event ID 1116 in PowerShell
6	Configure Wazuh to collect Defender Operational logs
7	Generate a fresh detection
8	Investigate the Wazuh SIEM alert
9	Document findings and analyst conclusion

10. Troubleshooting Performed

The Wazuh server IP changed after a VM restart, requiring the Windows agent manager address to be updated. The Windows Defender Operational event channel was also not initially present in the Wazuh agent configuration. After adding the eventchannel localfile entry and restarting the Wazuh service, Defender Event ID 1116 was successfully ingested.

11. Skills Demonstrated

Skill	Evidence
Malware triage	Reviewed file, hash, detection, severity, and endpoint context
Endpoint security	Used Microsoft Defender Protection History and Operational logs
Hashing	Calculated and validated a SHA-256 file fingerprint
Windows event analysis	Investigated Defender Event ID 1116
SIEM integration	Configured Wazuh to collect Defender Operational events
Threat hunting	Located and reviewed the Defender alert in Wazuh
Troubleshooting	Resolved agent IP/configuration and event ingestion issues

12. Conclusion

The lab successfully demonstrated a complete malware-detection and triage workflow using a safe EICAR test sample. Microsoft Defender detected and quarantined the file, generated Event ID 1116, and Wazuh ingested the event as a high-severity alert. The exercise demonstrated practical endpoint security monitoring, malware triage, Windows event analysis, SIEM integration, and troubleshooting.

Portfolio summary: Performed safe malware triage using EICAR, calculated a SHA-256 hash, validated Microsoft Defender detection and quarantine, configured Wazuh to ingest Defender Operational events, and investigated a high-severity Event ID 1116 alert.